
Lifecycle & Scheduling Tools in Kubernetes

These features allow fine-tuned control over how and when pods are scheduled, run, restarted, and terminated, ensuring reliability, performance, and efficient use of resources.

1. Init Containers

What they are:

Init containers are **special containers** that run **before** the main application containers in a pod.

Use Cases:

- Set up environment (e.g., copy config files, wait for a service)
- Perform initialization tasks like DB migrations
- Check dependencies

Characteristics:

- Run sequentially (one at a time)
- Must complete successfully for the main containers to start
- Defined in the `spec.initContainers` field

Example:

initContainers:

- name: init-db

image: busybox

command: ['sh', '-c', 'until nc -z db 5432; do sleep 1; done;']

2. Ephemeral Containers

What they are:

Temporary containers **added to running pods** for debugging purposes.

Use Case:

- Debug a running pod **without restarting** it
- Attach tools (like `curl`, `tcpdump`, `strace`) that are not in the original container image

Key Notes:

- Not restarted after exit
- Not part of the pod's lifecycle
- Cannot access volumes or be used for app logic

Usage:

```
kubectl debug -it <pod-name> --image=busybox
```

3. Affinity & Anti-Affinity Rules

What they do:

Control **where pods are scheduled** based on **node or pod attributes**.

Types:

- **Node Affinity** – Schedule pods on nodes with specific labels
- **Pod Affinity** – Schedule pods **near** other specific pods
- **Pod Anti-Affinity** – Schedule pods **away from** specific pods

Use Cases:

- Spread pods across zones

- Ensure certain services co-locate or avoid co-locating

Example (Node Affinity):

affinity:

nodeAffinity:

requiredDuringSchedulingIgnoredDuringExecution:

nodeSelectorTerms:

- matchExpressions:

- key: zone

operator: In

values:

- us-east-1a

Example (Pod Anti-Affinity):

podAntiAffinity:

requiredDuringSchedulingIgnoredDuringExecution:

- labelSelector:

matchExpressions:

- key: app

operator: In

values:

- backend

topologyKey: "kubernetes.io/hostname"

4. Taints and Tolerations

Taints (on nodes) prevent pods from being scheduled **unless they tolerate** those taints.

Use Cases:

- Reserve nodes for specific workloads (e.g., GPU, high-memory)
- Prevent noisy neighbor problems
- Isolate critical applications

How it works:

- **Taint** a node: prevents all pods from scheduling unless tolerated.
- **Toleration** on a pod: allows it to be scheduled onto tainted nodes.

Taint Example:

```
kubectl taint nodes node1 dedicated=ml:NoSchedule
```

Toleration Example (in pod spec):

tolerations:

```
- key: "dedicated"  
  operator: "Equal"  
  value: "ml"  
  effect: "NoSchedule"
```

5. PriorityClass

What it does:

Defines **priority levels** for pods, influencing:

- **Scheduling order** when resources are limited
- **Preemption**, where lower-priority pods are evicted to make room

Use Case:

- Critical system pods should run before user workloads
- Differentiate between batch and production workloads

Example PriorityClass:

```
apiVersion: scheduling.k8s.io/v1  
kind: PriorityClass  
metadata:
```

name: high-priority
value: 100000
globalDefault: false
description: "Used for critical workloads."

Then in the Pod:

priorityClassName: high-priority

🔧 6. PodDisruptionBudget (PDB)

What it does:

Limits the number of pods of a **replicated application** that can be down during voluntary disruptions (like upgrades or node maintenance).

Use Cases:

- Ensure availability during node upgrades or rolling updates
- Prevent entire replicaset from being disrupted

Specifies:

- **minAvailable**: Minimum number of pods that must be available
- **maxUnavailable**: Maximum number that can be unavailable

Example:

```
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: web-pdb
spec:
  minAvailable: 2
  selector:
    matchLabels:
      app: web
```

⚠️ 7. PodPreset (Deprecated)

What it was:

A way to **inject configuration (e.g., environment variables, volumes, volume mounts)** into pods at creation time.

Why it was deprecated:

- Complex and hard to manage/debug
- Admission webhooks now offer more flexibility and control

Deprecated since: Kubernetes v1.20+

Alternative: Use **mutating admission webhooks** or **sidecar injection**.

✅ Summary Table

Feature	Purpose	Still Used?
Init Containers	Setup tasks before main app container starts	✅ Yes
Ephemeral Containers	Debug live pods without restarts	✅ Yes
Affinity Rules	Schedule pods based on node/pod relationships	✅ Yes
Taints/Tolerations	Control which pods can run on which nodes	✅ Yes
PriorityClass	Control pod scheduling/preemption by importance	✅ Yes
PDB	Maintain availability during voluntary disruptions	✅ Yes
PodPreset	Inject config at pod creation time (now deprecated)	❌ No
